

# Danya Kogan

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## EDUCATION

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**Carnegie Mellon University** Pittsburgh, PA

*Bachelor of Science in Mechanical Engineering; Additional Major in Robotics, Minor in Anthropology. GPA: 3.72/4.0. May 2027*

## EXPERIENCE

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**Undergraduate Researcher, Zoe2 Field Rover** *Carnegie Mellon University Planetary Robotics Lab* March 2025 – Present

- Design, fabricate, and verify highly-critical components for a field robot using Solidworks CNC machining
- Assist in testing a driveable robot to gather data on robot performance

**Undergraduate Researcher** *Carnegie Mellon University Microrobotics Laboratory* October 2023 – December 2024

- Designed, documented, and constructed small-scale spring-based jumping robots using SolidWorks

## PROJECTS

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**Real-Time Transit Map** *Personal Project* May 2026 – Present

- Designing and fabricating a 450×450mm 3D-printed physical map with LED-illuminated, translucent road segments that display real-time bus positions using live transit API data
- Building Raspberry Pi software to convert live GPS coordinates into LED position updates; integrating 7-segment displays for arrival time estimates

**Zine Folding and Stapling Assist Tool** *Personal Project* December 2025 – Present

- Designed a knife-fold and stapling mechanism to semi-automate zine assembly using 3D printing, laser cutting, and hand tools, after an earlier buckle-fold approach proved unreliable in testing
- Iterating on stapler alignment and page-loading usability based on test-run data and user feedback; planning open-source release

**Cycloidal Gearbox** *Carnegie Mellon University Robotics Club, Mechanical Lead* March 2024 – April 2026

- Designed and fabricated a low-cost cycloidal gearbox from Delrin and steel using manual lathe, mill, and laser cutter for the Robotics Club's
- Designed and built custom test rigs to characterize dynamic and static torque performance
- Presented findings at CMU's Meeting of the Minds Undergraduate Research Symposium and documented manufacturing instructions via GitHub Wiki

**Soft Robotic Manipulator** *Personal Project* 2022 – 2023

- Designed a parametric silicone-finger mold and assembled multiple actuators into a soft manipulator
- Built and programmed an Arduino-based syringe pump to precisely control the manipulator pneumatically

## ACTIVITIES & LEADERSHIP

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**Carnegie Mellon Robotics Club** *Senior Member* Fall 2023 - Present

- Led quadruped robotics project as the mechanical lead

**Students for International Labor Solidarity** *Core Member* Fall 2024 – Present

- Facilitate weekly chapter meetings and coordinate with national organization leadership to align chapter activities with campaign strategy
- Organize and present public lectures on university apparel supply-chain transparency to the CMU student body as part of the "Follow the Thread" campaign
- Helped establish an ongoing working relationship with the university licensing director on supply-chain transparency, where none previously existed, through sustained engagement over multiple semesters

**README Campus Publication** January 2024 – Present

**Red Robot Hackathon 3rd Place and "Best Use of Rapid Prototyping" Award, presented by Caterpillar** October 2023

## SKILLS

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**CAD Software:** SolidWorks, Fusion 360, Onshape

**Programming:** Python, C++, Java, MATLAB, JavaScript, Arduino, CircuitPython, SQL, ROS

**Machines:** Manual Mill, CNC Mill, CNC Router, Manual Lathe, Laser Cutter, FDM 3D Printer, SLA/DLP 3D Printer

**Miscellaneous Technical:** Linux, Microcontrollers, FEA, Ansys, MasterCAM, Fusion 360 CAM